

Econ 280 - Computation

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1 Extension

This is an extension on the paper Risk Pooling, Risk Preferences, and Social Networks by Attanasio et al (American Economic Journal: Applied Economics 2012). I examine the heterogeneity in defection behavior by introducing a Gender Interaction Specification and performing a robustness check with Municipality Fixed Effects (FE). In the context of the paper, defection in a dyad refers to an individual's choice to leave a risk-sharing group after observing their own gamble outcomes in the second round of the experiment. While each participant plays their gamble game independently, forming a group creates an implicit agreement to share their earnings equally within the group. However, since group agreements are unenforced, a participant can defect by choosing not to share their winnings while retaining their own gains. This defection behavior highlights the tension between trust and opportunism in the group setting. Defection disrupts the intended mutual insurance mechanism of the risk-sharing group and reflects the individual's lack of commitment to the group agreement. The estimating equation for the Gender Interaction Specification includes an interaction term between the genders of the two players to capture gender-specific dynamics:

$$Defection_{ij} = \beta_0 + \beta_1 GambleDiff_{ij} + \beta_2 (FemaleA_{ij} \times FemaleB_{ij}) + \mathbf{X}_{ij}\beta + \epsilon_{ij}, \quad (1)$$

where $Defection_{ij}$ represents the likelihood that individual i defects from the risk-sharing group formed with individual j . The intercept, β_0 , captures the baseline likelihood of defection when all other predictors are zero. The variable $GambleDiff_{ij}$, with coefficient β_1 , measures the difference in gamble choices between the two individuals, capturing the impact of risk preference mismatches on defection behavior. The vector $\mathbf{X}_{ij}$ includes control variables such as differences in age, education, and income, as well as indicators for social ties (friendship or family) and the individual genders of Player A and Player B. The interaction term, $(Female_i \times Female_j)$, with coefficient β_2 , accounts for the effect of the joint gender composition of the dyad on defection behavior. Finally, ϵ_{ij} represents the error term, capturing unobserved factors influencing the likelihood of defection. The robustness check incorporates municipality-level fixed effects to account for unobservable regional factors.

Table 1: Marginal Effects of Defection Behavior in Dyad Analysis

	(1) Naive	(2) Gender Interaction	(3) Municipality FE
Difference in Round 1 Gamble Choice	0.0308*** (0.00960)	0.0308*** (0.00960)	0.0366*** (0.00997)
Difference in Age	-0.00139 (0.00129)	-0.00136 (0.00129)	-0.00126 (0.00135)
Difference in Years of Schooling	-0.00858* (0.00450)	-0.00858* (0.00450)	-0.00430 (0.00468)
Difference in Income	0.00843 (0.0250)	0.00855 (0.0250)	-0.0231 (0.0267)
One Recognised Friendship, Other Family Tie	0.265 (0.265)	0.265 (0.265)	0.164 (0.274)
Player A Female	-0.0125 (0.0373)	-0.0566 (0.0863)	0.0296 (0.0887)
Player B Female	-0.121*** (0.0362)	-0.166* (0.0870)	-0.0850 (0.0894)
Gender Interaction		0.0542 (0.0957)	0.0991 (0.0981)
Observations	88266	88266	73860

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Across all specifications, the marginal effect of differences in Round 1 gamble choices consistently shows a significant impact on defection behavior. For example, a one-unit increase in the difference in gamble choices between two players increases the likelihood of defection by approximately 3.1 percentage points in the Naive Specification. This effect remains stable in the Gender Interaction Specification (3.1 percentage points) and even strengthens slightly in the Municipality FE Specification (3.7 percentage points), indicating robust predictive power. These results support the central role of risk preference mismatches in shaping defection decisions.

Demographic variables, including differences in age and schooling, exhibit smaller and less consistent effects. For instance, the marginal effect of a one-year difference in schooling reduces the likelihood of defection by 0.86 percentage points in the Naive Specification, but this effect diminishes in magnitude and significance when fixed effects are included. Similarly, differences in income show no significant predictive power in the Naive or Gender Interaction Specifications and even shift direction in the Municipality FE Specification (-2.3 percentage points).

The introduction of the Gender Interaction Specification reveals nuanced dynamics between the genders of the two players. Player B's gender is consistently significant, with female players reducing defection likelihood by 12.1 percentage points in the Naive Specification and 16.6 percentage points in the Gender Interaction Specification. However, the interaction term between the genders of the two players is not statistically significant, suggesting limited evidence for combined gender effects. This finding implies that the role of gender may be more individual-specific than relational in this context.

The Municipality FE Specification provides additional robustness to the analysis by accounting for regional heterogeneity. The predictive power of gamble choice differences remains strong and statistically significant.